

Richard Oliveira

Doctoral Researcher at Politecnico di Milano

[Google Scholar](#) [LinkedIn](#) [GitHub](#) [Website](#)

I'm a [Marie Skłodowska-Curie](#) Doctoral fellow working on distributed optimization and machine learning theory through the lens of applied harmonic analysis and signal processing.

Education

Politecnico di Milano

2025 - Present

▷ PHD IN DATA ANALYTICS & DECISION SCIENCES.
Advised by [Dr. Stefano Savazzi](#) and [Prof. Monica Nicoli](#).

University of California, San Diego

2022 - 2024

▷ MSC IN ELECTRICAL ENGINEERING, SIGNAL & IMAGE PROCESSING.
Advised by [Prof. Massimo Franceschetti](#).

2020 - 2022

▷ BSC IN ELECTRICAL ENGINEERING.
Advised by [Prof. Nick Antipa](#).

Research Experience

Politecnico di Milano

Electronics, Information Technology, and Bioengineering Department

2025 - Present

▷ Doctoral Researcher.
Advisors: [Dr. Stefano Savazzi](#) and [Prof. Monica Nicoli](#).
Distributed optimization theory and mathematical foundations of agentic systems.

University of California, San Diego

Electrical and Computer Engineering Department

2023 - 2024

▷ Graduate Student Researcher.
Advisor: [Prof. Massimo Franceschetti](#).

Contributed to the theory of discrete Pearson-Rayleigh random walks by analyzing how walks of this kind behave in discrete space with geometric and Poisson distributed step sizes.

2020 - 2022

▷ Undergraduate Student Researcher.
Advisor: [Prof. Nick Antipa](#).
Analysis and development of a linear, spatial-invariant imaging system, DiffuserCam; experimental recovery of a point spread function; Python implementation of an accelerated ADMM solver for inverse-problem scene reconstruction.

Publications

Journal Articles

Feb. 2026

▷ ON SOLUTIONS TO DISCRETE-TIME INVERSE PROBLEMS VIA SPARSE SUPERPOSITIONS OF DECAYING SHIFTED HEAVISIDE FUNCTIONS: A HARDY SPACE APPROACH
[R. Oliveira](#) and [S. Savazzi](#). *IEEE Signal Processing Letters*.

Apr. 2025

▷ DISTRIBUTED COMPOSITE REGULARIZATION VIA DUAL-LAYER CONSENSUS OPTIMIZATION
[R. Oliveira](#) and [S. Savazzi](#). *IEEE Transactions on Signal and Information Processing over Networks*.

July 2026

▷ ON THE STABILITY OF CONSENSUS GRADIENT DYNAMICS FOR REGULARIZED DISTRIBUTED DEEP MATRIX FACTORIZATION: EXPLICIT BOUNDS
[R. Oliveira](#) and [S. Savazzi](#). *IEEE Signal Processing Letters*.

Manuscripts Under Review

June. 2026

▷ DEEP NEURAL NETWORKS WITH EXPONENTIALLY DAMPED RIDGE SPLINE ACTIVATIONS: A FUNCTIONAL-ANALYTIC APPROACH TO DISTRIBUTED LEARNING
[R. Oliveira](#), [S. Savazzi](#) and [M. Nicoli](#). Submitted to *IEEE Transactions on Neural Networks and Learning Systems*.

Projects

Nov. - Dec. 2024

▷ **Inverse Problems in NeuroMANCER**
Used NeuroMANCER to solve an MNIST inverse problem with physics-informed constraints and non-smooth regularizers. Compared results against CVXPY and identified the careful fine-tuning required for non-differentiable regularization.

Mar. - Jun. 2024

▷ **Deep Pose Estimation from 3D Point Clouds**
Developed a neural network for relative-pose prediction between 3D point-cloud sets from LiDAR scans and RGBD camera data from the EDEN dataset, addressing initialization sensitivity and the computational cost of repeated alignment in classical registration methods.

- Jan. - Mar. 2024 ▷ **Visual-Inertial SLAM**
Implemented a SLAM pipeline using the Extended Kalman Filter: prediction for localization, updates for landmark mapping, and the combined prediction-update loop for simultaneous localization and mapping.
- Jan. - Mar. 2024 ▷ **LiDAR-based SLAM**
Approached SLAM for a differential-drive robot using wheel encoders, IMU, LiDAR, and RGBD camera measurements. Used ICP and GTSAM for trajectory estimation and constructed an occupancy grid from the optimized trajectory and LiDAR measurements.
- Jan. - Mar. 2023 ▷ **Theory and Applications of Convex Lifts**
Prepared a literature review on lifts of convex sets, emphasizing how higher-dimensional convex representations can simplify structure and support optimization methods.
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Teaching

University of California, San Diego

Electrical and Computer Engineering Department

- Mar. - Jun. 2024 ▷ Graduate Teaching Assistant, ECE 227: Big Network Data.
Developed robotics-oriented course projects for a course on graph analysis; provided feedback and instructional support during lecture hours.
- Sep. - Dec. 2021 ▷ Undergraduate Teaching Assistant, ECE 35: Introduction to Analog Design.
Provided academic support during lectures, developed instructional materials, attended lab sessions, assisted in exam grading, and held weekly office hours.
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Employment

Italian National Research Council (CNR)

Institute of Electronics, Computer and Telecommunication Engineering (IEIIT), Milano

- 2025 - Present ▷ Doctoral Researcher.
Distributed optimization theory, regularization, consensus networks and learning theory.

Onikron LLC

Remote

- 2024 - 2025 ▷ Junior Data Scientist.
Implemented and researched modern NLP models, including transformers and RNNs in PyTorch; developed lemmatization and tokenization scripts for data preprocessing; worked on clustering for unsupervised learning.

General Atomics A.S.I.

San Diego, CA

- Jun. - Aug. 2023 ▷ Algorithms and Signal Processing Engineer.
Researched modern neural-network architectures for computer-vision deep learning; created a Python/Bokeh data-visualization tool for heat-map analysis and preprocessing.

Ciena

San Jose, CA

- May - Aug. 2021 ▷ Database Engineer.
Used SQL queries through the FLYBID database to organize sensitive company records; focused on data dictionaries and LucidChart-based data mapping.
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Selected Honors & Awards

- 2025 - 2028 ▷ **Marie Skłodowska-Curie Doctoral Fellowship.**
- Jan. - Mar. 2024 ▷ **Graduate Student Researcher Grant.**
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Skills

Programming
Languages

- ▷ C, C++, MATLAB, SQL, Python, PyTorch, TensorFlow, Assembly, Scikit-learn.
▷ Italian; English.